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Postprandial Intestinal Blood Flow, Metabolic Rates, and Exercise in Chinook Salmon (*Oncorhynchus tshawytscha*)

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ABSTRACT

Following a relatively large meal (2% body mass of dry pellets), intestinal blood flow in chinook salmon (*Oncorhynchus tshawytscha*) increased significantly, up to 81%, between 14 and 29 h postprandially. Also, 15 h postprandially, oxygen consumption ($\dot{M}O_2$) was elevated by 128% compared with a measurement of routine $\dot{M}O_2$ made after 1 wk of fasting. The postprandial increase in $\dot{M}O_2$ (the heat increment) was $33 \mu\text{mol O}_2 \text{ min}^{-1} \text{ kg}^{-1}$. Because intestinal blood flow is known to decrease during swimming activity in fish, we therefore tested the hypothesis that swimming fish would have to make a trade-off between maximum swimming activity and digestive activity by comparing the swimming performance and metabolic rates of fed and fasted chinook salmon. As expected, $\dot{M}O_2$ increased exponentially with swimming velocity in both fed and fasted fish. Moreover, the heat increment was irreducible during swimming, such that $\dot{M}O_2$ remained approximately $39 \mu\text{mol O}_2 \text{ min}^{-1} \text{ kg}^{-1}$ higher in fed fish than in fasted fish at all comparable swimming speeds. However, maximum $\dot{M}O_2$ was unaffected by feeding and was identical in both fed and fasted fish ($\sim 250 \mu\text{mol O}_2 \text{ min}^{-1} \text{ kg}^{-1}$), and, as a result, the critical swimming speed (U_{crit}) was 9% lower in the fed fish. Three days after the fish were fed and digestion was completed, $\dot{M}O_2$ and U_{crit} were not significantly different from those measured in fasted fish. The ability of salmonids to maintain feeding metabolism during prolonged swimming performance is discussed, and it is suggested that reduced swimming performance may be due to postprandial sparing of intestinal blood to support digestion, thereby limiting the allocation of blood flow to locomotory muscles.

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Introduction

The total metabolism (as indicated by oxygen consumption: $\dot{M}O_2$) of animals is commonly described in terms of the costs of maintenance, digestion and assimilation, and locomotion.

The heat increment (HI; also termed specific dynamic action; Beamish and Trippel 1990), which is the combined mechanical and biochemical metabolic cost of digestion and assimilation, is attributed to a number of factors, such as the mechanical cost of moving the digesta through the intestines, the cost of digestion and absorption, and the assimilation of macromolecules mainly as protein synthesis. Tandler and Beamish (1979) estimated that the “mechanical component” of the HI was 10%–30%, depending on ration. However, Jobling and Davies (1980) were unable to detect any metabolic cost of intestinal peristalsis, suggesting that normally this component of digestion is small. Instead, most of the increase in $\dot{M}O_2$ appears to be due to processes that occur after nutrients have been taken up into the bloodstream. In fact, an increase in $\dot{M}O_2$, similar to the HI, can be initiated by intra-arterial injections of amino acids (Coulson et al. 1978; Brown and Cameron 1991). The metabolic cost of excreting nitrogenous products is small in salmonids (Randall and Wright 1987), since most of the nitrogen is excreted in the form of ammonia rather than urea (Brett and Zala 1975). Therefore, a large fraction of the HI is due to a more rapid turnover of proteins, which is observed in most tissues following feeding (McMillan and Houlihan 1988; Brown and Cameron 1991; Lyndon et al. 1992).

Most metabolic studies have considered the metabolic cost of either the HI or locomotion. More recently, there has been interest in the combined effects of HI and locomotion, that is, postprandial exercise. In nature, animals need to feed and exercise to varying degrees, and their ability to do both activities simultaneously could have a significant bearing on survival. For example, Alsop and Wood (1997) reported a 60% increase in metabolic rate in juvenile (5–10 g) rainbow trout (*Oncorhynchus mykiss*) fed to satiation, and this elevated metabolic rate was maintained throughout a critical swimming speed (U_{crit}) test, when $\dot{M}O_2$ increased exponentially as swimming speed increased. However, the fed fish ended up having a 15% lower U_{crit} and a similar maximum $\dot{M}O_2$ compared with fasted, control fish. Similarly, Beamish (1974) observed that the HI of large-mouth bass was unchanged up to a velocity approaching

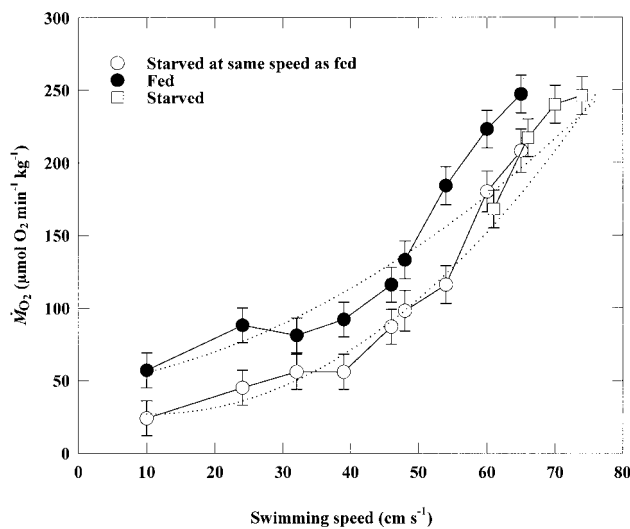


Figure 1. Oxygen consumption ($\dot{M}_{O_2}$) of 11 chinook salmon swimming first starved (open symbols) and then 15 h postprandially (filled symbols). The data were analyzed in several sections. First, the mean $\dot{M}_{O_2}$ of fed (filled circles) and starved fish (open circles) swimming at the five lowest swimming velocities were compared. Then the $\dot{M}_{O_2}$ of fed fish swimming at the last four highest velocities before they fatigued were compared with that of starved fish swimming at the same speed. The $\dot{M}_{O_2}$ of unfed fish swimming at the four highest velocities before exhaustion is also shown (open squares). The $\dot{M}_{O_2}$ of fed fish were significantly ($P < 0.05$) higher than in the starved fish, but $\dot{M}_{O_{2max}}$ were not significantly different in the two groups. The dotted lines show polynomials fitted to the data for fed and starved fish.

$\dot{M}_{O_{2max}}$, although his fish were not swum to U_{crit} . In contrast to rainbow trout, the $\dot{M}_{O_{2max}}$ of cod (*Gadus morhua*) swimming postprandially appears to be higher than in unfed fish (Blaikie and Kerr 1996). This suggests that cod may be able to support at least a part of their HI while swimming at close to 66% of U_{crit} . Even so, it is not known whether U_{crit} of cod is compromised postprandially. Also, in monitor lizards (*Varanus exanthematicus*), an 18% postprandial increase in $\dot{M}_{O_2}$ at rest was maintained at all levels of activity. But in contrast to juvenile rainbow trout, postprandial lizards had a higher maximum $\dot{M}_{O_2}$ compared with fasted lizards, and maximum walking speed was not compromised (Bennett and Hicks 2001). Similarly, the $\dot{M}_{O_{2max}}$ is higher in fed than in fasted Burmese pythons (Secor et al. 2000).

What emerges from these studies are two models for the effects of feeding on exercise. On the one hand are the juvenile rainbow trout that appear to prioritize oxygen delivery to maintain the HI during maximal prolonged swimming activity. On the other hand, Atlantic cod and reptiles such as monitor lizards and pythons can add the HI onto the cost of activity. These apparently contrasting results may reflect species differences in cardiorespiratory design with respect to oxygen delivery to tissues. Therefore, we undertook a study with chinook salmon,

which have a larger arterial oxygen transport capacity ($\dot{V}_{O_{2max}} = \text{maximum cardiac output} \times \text{oxygen concentration of arterial blood}$) that is much higher than that of rainbow trout (Kiceniuk and Jones 1977; Thorarensen et al. 1993; Gallagher et al. 1995, 2001).

The $\dot{M}_{O_{2max}}$ of fish appears to be primarily determined by the $\dot{V}_{O_{2max}}$ (Gallaugh et al. 2001). Therefore, the degree to which both the HI and activity can be maintained will depend on the balance between the $\dot{V}_{O_{2max}}$ and the aerobic demands of the HI and the locomotory muscles. We suggest that the $\dot{V}_{O_{2max}}$ of monitor lizards and Burmese pythons is high enough to accommodate both the requirements of the HI and activity, whereas the cardiovascular system of rainbow trout cannot support both functions simultaneously. Consequently, we anticipated that the high $\dot{V}_{O_{2max}}$ of chinook salmon would support the HI while swimming postprandially. Moreover, postprandial intestinal blood flow was measured to assess the potential for blood flow redistribution during swimming.

Material and Methods

Experimental Animals

Chinook salmon (mean mass $\pm$ SEM = 520 ± 30 g; mean length = 33.7 ± 0.5 cm) used for this experiment were raised from juveniles at the West Vancouver Laboratory of Fisheries and Oceans, West Vancouver, British Columbia. They were raised under natural photoperiod in outdoor tanks and provided with an ample supply of seawater ($9^\circ\text{--}10^\circ\text{C}$; salinity $30\text{‰} \pm 1\text{‰}$). The fish were fed dry pellets (Pacific Aquafoods, Vancouver, British Columbia) three times a week.

Postprandial Exercise Protocol

Before the swim trials, the fish ($N = 11$) were fasted for 1 wk and transferred to a Brett-type swim tunnel, where they were allowed to habituate for 15 h before swimming to U_{crit} . This is the same procedure as used by Thorarensen et al. (1993) and Gallagher et al. (1995, 2001) and appears to give salmonids ample time for recovery. Resting $\dot{M}_{O_2}$ was measured for fasted fish before U_{crit} and $\dot{M}_{O_2}$ were determined in velocity increment tests, as described by Thorarensen et al. (1993). Briefly, swimming speed was increased first to 0.75 body length s^{-1} and then in steps of 0.25 body length s^{-1} . The duration of each step was 20 min or until the fish fatigued. $\dot{M}_{O_2}$ was measured at all velocity increments, including the increment just before fatigue, where maximum $\dot{M}_{O_2}$ ($\dot{M}_{O_{2max}}$) was attained. After the initial test, fish were returned to the holding tank. Two days later, fish were netted again, lightly anesthetized (2-phenoxyethanol), and force-fed a paste made from equal volumes of dry pellets (2% body mass) and water. The paste was administered by gavage into the stomach through a plastic tube. The fish were placed in the swim tunnel, and U_{crit} and $\dot{M}_{O_{2max}}$ were determined 15 h postprandially. This time was chosen because other studies

Table 1: Oxygen consumption ($\dot{M}O_2$) and swimming performance (U_{crit}) of 11 chinook salmon swimming after 1 wk starvation, 15 h postprandially, and 63 h postprandially

Condition	$\dot{M}O_2$ ($\mu\text{mol O}_2 \text{ min}^{-1} \text{ kg}^{-1}$)		U_{crit}	
	Rest	Maximum	Body Length s^{-1}	cm s^{-1}
Fasted	25 ± 3^A	255 ± 11	$2.2 \pm .1^A$	74.6 ± 2.6^A
15 h postprandially	57 ± 3^B	249 ± 11	$2.0 \pm .1^B$	67.6 ± 3.5^B
63 h postprandially	29 ± 3^A	266 ± 11	$2.2 \pm .1^{AB}$	73.4 ± 3.7^{AB}

Note. Mean values $\pm$ SEM. Variables denoted with different letters are significantly different ($P < 0.05$; Tukey's test).

have indicated that HI has reached a maximum at 10–15 h postfeeding (Saunders 1963; Muir and Niimi 1972; Legrow and Beamish 1986; Lyndon et al. 1992; see also Fig. 1). The fish were never seen regurgitating the food, and therefore it was assumed that it was digested. A third measurement of U_{crit} and $\dot{M}O_{2max}$ was performed 48 h after the second swim (63 h postprandially). The fasted fish were lightly anesthetized and “sham-fed” by inserting a tube into the stomach before they were placed in the swim tunnel to recover for 15 h before swimming. The same procedure was also used in the initial trial of fasted fish.

Postprandial Intestinal Blood Flow Protocol

The fish used in this study (body mass 200–350 g) were also raised from juveniles at the West Vancouver Laboratory of Fisheries and Oceans and from the same stock used in the previous experiment as well as by Thorarensen et al. (1993). They were also reared under similar conditions, except ambient temperature in this study was 12°–13°C.

A Doppler flow probe was surgically placed on the intestinal aorta of anesthetized fish through a lateral incision, as described by Thorarensen et al. (1993). The fish were not fed for 24 h before the surgery, and during the experiment they were contained in black holding boxes. After a 24-h recovery period, blood flow in the intestinal artery ($\dot{q}_{IA}$) was recorded for 10 min h^{-1} for 24 h. The recordings during the last 5 h were averaged and presented as the resting value. Eight fish were then force-fed, following the same procedure as described above, and replaced in the holding box. Blood flow in the intestinal artery was then monitored, as before, for 40 h. Control fish ($N = 5$) were sham-fed, as described above, and treated identically to fed fish, except food was not introduced to the stomach.

Statistics

The $\dot{M}O_{2max}$ and U_{crit} of fed and fasted fish were compared with a two-way ANOVA (individual fish $\times$ swim trial). Polynomial models were fitted to the $\dot{M}O_2$ data at different swimming velocities, as described by Thorarensen et al. (1993). However,

the polynomial lines did not describe the original data faithfully for fed fish swimming at high velocities, where observed values tended to be higher than predicted values (Fig. 1). Therefore, the $\dot{M}O_2$ data for fed and starved fish swimming at identical velocities were analyzed by comparing with ANOVA, and because of variability in swimming performance, the data were compared in two parts. First, $\dot{M}O_2$ values at the five lowest swimming velocities were compared for fed and starved fish. Second, $\dot{M}O_2$ values of fed fish swimming at the four highest swimming velocities were compared with the $\dot{M}O_2$ values of fasted fish swimming at the same velocities. The different data analysis sections are separated by different connecting lines in Figure 1. Finally, the mean $\dot{M}O_2$ of the fasted fish swimming at the four highest swimming velocities was calculated (Fig. 1,

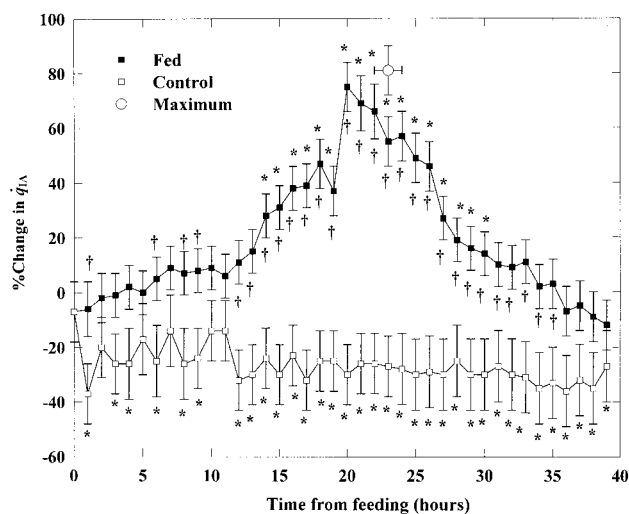


Figure 2. Postprandial increase in intestinal blood flow ($\dot{q}_{IA}$) of chinook salmon ($N = 8$) force-fed a meal of paste made from equal volumes of dry feed (2% body mass) and water. Control fish ($N = 5$) were sham-fed (open symbols). The mean value and average timing for the peak increase in $\dot{q}_{IA}$ for individual fish is indicated by the open circle. Vertical and horizontal bars represent SEM. Asterisks denote a statistically significant difference ($P < 0.05$) from the initial prefeeding level, as indicated by a Tukey's test. Daggers denote a statistically significant difference ($P < 0.05$) between corresponding values for fed and control fish.

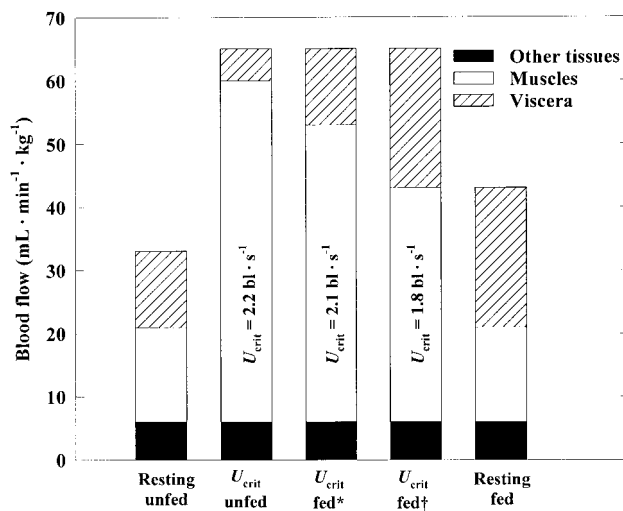


Figure 3. Blood flow distribution in unfed and fed chinook salmon either resting or swimming. The numbers on the columns indicate the estimated U_{crit} , assuming that the cost of swimming increases with velocity squared and is proportional to the estimated muscle blood flow. The blood flow distribution is based on direct measurements of gut blood flow in chinook salmon (Thorarensen et al. 1993) and estimates of blood flow distribution with injections of radiolabeled microspheres (Thorarensen 1994). Asterisk indicates gut blood flow the same as in unfed resting fish. Dagger indicates gut blood flow the same as in fed resting fish.

open squares). The resting and maximum $\dot{M}O_2$ of fish swimming unfed, 15 h postprandially, and 63 h postprandially were compared with one-way ANOVA, and where a significant difference was found, the mean $\dot{M}O_2$ in different trials was compared with Tukey's tests.

Results

Postprandial Exercise

$\dot{M}O_2$ increased 10-fold in fasted fish during the swim to U_{crit} (Fig. 1). Their aerobic scope was $230 \mu\text{mol O}_2 \text{ min}^{-1} \text{ kg}^{-1}$. The HI elevated routine $\dot{M}O_2$ by $33 \mu\text{mol O}_2 \text{ min}^{-1} \text{ kg}^{-1}$ at 15 h postprandially (Table 1). Thus, the HI represented a 128% increase over routine $\dot{M}O_2$ and nearly 15% of the aerobic scope. Moreover, $\dot{M}O_2$ of fed fish remained, on average, $39 \mu\text{mol O}_2 \text{ min}^{-1} \text{ kg}^{-1}$ higher ($P < 0.05$) than that of fasted fish at all swimming velocities, including U_{crit} . Therefore, HI did not change significantly with increasing swimming velocities (Fig. 1). In addition, because $\dot{M}O_{2max}$ was not significantly different in fed and fasted fish, U_{crit} for fed fish was significantly ($P < 0.05$) 9% lower (Table 1) than that of fasted fish.

At 63 h postprandially, U_{crit} and the $\dot{M}O_{2max}$ were not significantly different from values for fasted fish (Table 1). This finding indicates that repeated swimming trials did not affect performance and that most of the metabolic consequences of the food ration had subsided.

Postprandial Intestinal Blood Flow

Chinook salmon fed a meal of 2% body mass increased $\dot{q}_{IA}$ significantly from 14 h postprandially through to 29 h postprandially (Fig. 2). At 20 h postprandially, $\dot{q}_{IA}$ had increased by $78\% \pm 9\%$. When peak flows for individual fish were averaged, peak $\dot{q}_{IA}$ occurred at 23 ± 1 h postprandially and represented an $81\% \pm 9\%$ increase (Fig. 2). One hour after sham-feeding, the fasted fish experienced a $37\% \pm 11\%$ decrease in $\dot{q}_{IA}$. After that, there was no significant change in $\dot{q}_{IA}$, which remained on average reduced by 26% compared with pretreatment control (Fig. 2).

Discussion

We have demonstrated that, when faced with the dual challenge of swimming and digestion, chinook salmon will give preference to the HI at the expense of maximum activity levels. In fact, HI remained remarkably fixed at all swimming velocities, suggesting that the metabolic processes behind the HI were not compromised. Instead, less oxygen was made available for locomotion since the maximum swimming performance (U_{crit}) was reduced by 9%. These results are in accord with those reported for juvenile rainbow trout (Alsop and Wood 1997), where the HI was also irreducible during swimming. Similarly, they also found that the $\dot{M}O_{2max}$ of juvenile rainbow trout was not significantly different in fed and fasted fish, while the U_{crit} of fed fish was decreased by 15% (Alsop and Wood 1997). Thus, neither juvenile rainbow trout nor larger, more athletic chinook salmon could simultaneously support both the oxygen demands of the HI and the maximum demands of the locomotory muscles. These findings are in line with the $\dot{M}O_{2max}$ of salmonids being determined by oxygen delivery rather than the maximum oxygen demands of all tissues.

Given that there is mounting evidence to suggest that the $\dot{M}O_{2max}$ of salmonids is primarily limited by the oxygen-transporting capacity of the cardiovascular system (Gallaughier et al. 1995, 2001), it is possible to examine these results in the context of a potential limit on blood flow distribution when chinook salmon feed and swim; that is, $\dot{T}O_{2max}$ of chinook salmon is insufficient to simultaneously support the maximum oxygen demands of the HI and the locomotory muscles. Following a meal, $\dot{q}_{IA}$ in fish is known to increase by 40%–100% to facilitate absorption and support the metabolic demands of the viscera (Axelsson and Fritzsche 1991; Axelsson et al. 1989, 2000). Our results of an 81% postprandial increase in $\dot{q}_{IA}$ are consistent with these earlier works. Also, in unfed salmonids, increasing swimming velocity from rest to maximum elevates $\dot{Q}$ (Kiceniuk and Jones 1977; Thorarensen et al. 1996; Gallaughier et al. 2001), increases perfusion of locomotory muscles, and decreases blood flow to the gut (Randall and Daxboeck 1982; Thorarensen et al. 1993). Gut blood flow has not been measured directly in fed, swimming fish; however, it has been suggested that blood flow redistributed from the intestines constitutes a significant

Table 2: Resting and maximum oxygen consumption ($\mu\text{mol O}_2 \text{ min}^{-1} \text{ kg}^{-1}$) in five species of ectotherms while either fed or unfed

	Rainbow Trout (<i>Oncorhynchus mykiss</i>) ^a	Cod (<i>Gadus morhua</i>) ^b	Chinook Salmon (<i>Oncorhynchus tshawytscha</i>) ^c	Monitor Lizard (<i>Varanus exanthematicus</i>) ^d	Burmese Python (<i>Python molurus</i>) ^e
Unfed:					
$\dot{M}\text{O}_{2\text{rest}}$	42	42	25	89	34
$\dot{M}\text{O}_{2\text{max}}$	167	81	255	710	339
Fed:					
$\dot{M}\text{O}_{2\text{rest}}$	67	62	57	183	223
$\dot{M}\text{O}_{2\text{max}}$	167	94 ^f	249	839	397
Scope for activity unfed	125	39	230	621	305
Scope for activity fed	100	32	192	656	174

^a Alsop and Wood 1997.^b Blaikie and Kerr 1996.^c This study.^d Bennett and Hicks 2001.^e Secor et al. 2000.^f Fish were swimming at 66% of U_{crit} .

portion of the increased blood flow to locomotory muscles in unfed salmonids (Thorarensen et al. 1993; Thorarensen 1994; Gallagher et al. 2001). We propose that the conflicting requirements for blood flow to the intestines and the locomotory muscles may compromise the ability of salmonids to simultaneously swim and digest and that, in fed fish, little if any blood flow is diverted from the viscera to the locomotory muscles, resulting in reduced swimming performance of fed fish compared with unfed fish.

It is impossible to directly measure the total blood flow to the locomotory muscles, and therefore it is difficult to test the above hypothesis without relying on estimates of skeletal muscle blood flow using microsphere techniques. Here, we have estimated (Fig. 3) the contribution of blood flow redistribution from the viscera to the swimming performance of chinook salmon on the basis of data from this experiment and earlier studies (Thorarensen et al. 1993; Thorarensen 1994; Gallagher et al. 2001). In resting unfed fish, blood flow to the intestines is $12 \text{ mL min}^{-1} \text{ kg}^{-1}$ (Thorarensen et al. 1993), blood flow to the muscles is $15 \text{ mL min}^{-1} \text{ kg}^{-1}$, and other tissues receive $6 \text{ mL min}^{-1} \text{ kg}^{-1}$. At swimming velocities close to U_{crit} , blood flow to the locomotory muscles of unfed chinook salmon can reach $54 \text{ mL min}^{-1} \text{ kg}^{-1}$ (Thorarensen 1994), and in resting and fasted chinook salmon, $\dot{q}_{\text{IA}}$ is about $12 \text{ mL min}^{-1} \text{ kg}^{-1}$ (Thorarensen et al. 1993). For simplicity, we assume that blood flow to other tissues remains fixed at $6 \text{ mL min}^{-1} \text{ kg}^{-1}$ regardless of activity or feeding status, although arguments could also be made for increases and decreases. Unfed chinook salmon swimming at U_{crit} reduce $\dot{q}_{\text{IA}}$ by 60%–70% (Thorarensen et al. 1993), thus potentially providing $7 \text{ mL min}^{-1} \text{ kg}^{-1}$ for perfusion of locomotory muscles (i.e., 13% of the locomotory muscle's needs; Fig. 3). The postprandial increase in $\dot{q}_{\text{IA}}$ was 81% (Fig. 2), reaching a maximum of $22 \text{ mL min}^{-1} \text{ kg}^{-1}$, which if unchanged during swimming would represent a "stealing" of $18 \text{ mL min}^{-1} \text{ kg}^{-1}$ from other tissues, assuming that $\dot{T}\text{O}_{2\text{max}}$ cannot

be increased in chinook salmon, as appears to be the case here. If it is further assumed that the cost of swimming increases with velocity squared (Thorarensen et al. 1993) and that U_{crit} is limited by the oxygen supply to the muscles, we can estimate the change in U_{crit} if muscle blood flow was reduced. Thus, U_{crit} would be reduced by 7% (to $2.1 \text{ body length s}^{-1}$) in unfed fish if no blood flow was redistributed to the muscles and by as much as 17% (to $1.81 \text{ body length s}^{-1}$) if visceral blood flow was maintained at maximum postprandial levels (Fig. 3). These rather simple predictive calculations show that the shifts in blood flow between the viscera and the locomotory muscles are of a magnitude that could account for the observed 9% reduction in U_{crit} of fed fish.

Ultimately, whether an animal can simultaneously digest and exercise will depend on the absolute rather than relative values of aerobic scope, HI, and $\dot{M}\text{O}_{2\text{max}}$. Therefore, given the apparent difference between Atlantic cod and salmonids in this regard, it is instructive to compare these species (Table 2). The aerobic scope of cod is much lower than that of chinook salmon (Table 2), and the U_{crit} of cod is lower than that of chinook salmon (Reidy et al. 2000). Even when swimming at 67% of U_{crit} , the $\dot{M}\text{O}_2$ of cod was not significantly different from rest (Blaikie and Kerr 1996). However, Atlantic cod, like salmonids, maintain their HI while swimming at 67% of U_{crit} ; in fact, the HI appears to increase at higher swimming velocities (Blaikie and Kerr 1996). Therefore, the $\dot{M}\text{O}_2$ of fed fish swimming at 67% of U_{crit} was higher than that of unfed fish swimming at U_{crit} (Blaikie and Kerr 1996). Also, the $\dot{M}\text{O}_{2\text{max}}$ of Atlantic cod recovering from bouts of intensive exercise may also be higher than when measured at U_{crit} (Soofiani and Priede 1985). These results suggest that in contrast to salmonids, the $\dot{T}\text{O}_{2\text{max}}$ of Atlantic cod can support oxygen demands in excess of those of the locomotory muscles. However, since Blaikie and Kerr (1996) did not swim the fed fish up to U_{crit} , it is not known

whether, as for salmonids, the maximum swimming performance of Atlantic cod is limited after feeding.

Furnell (1987) reported that the HI of sablefish was reduced as swimming speed increased and suggested that sablefish had a physiological mechanism to reduce O_2 allocation to digestive functions. This may represent a species difference between sablefish, on the one hand, and chinook salmon, rainbow trout, Atlantic cod, and largemouth bass, on the other. However, close inspection of Furnell's results suggests that this is not necessarily the case. First, Furnell studied only three fish, and they were not even swum to U_{crit} . Second, log-linear lines were fitted to the data. The results from this study suggest that, in some cases, neither exponential nor polynomial models can faithfully reproduce the original data. In fact, when polynomial models were fitted to our data (Fig. 1), the curves for fed and starved fish converged in a manner similar to that observed by Furnell (1987) but diverged from the original data at the highest velocities, and we therefore rejected this statistical approach. In Furnell's study, there were only a few data points for $\dot{M}O_2$ at velocities where the curves converged. Therefore, his conclusion that the curves converge is largely based on the projection of fitted lines leading to uncertainty regarding the results for the sablefish.

It should be kept in mind that the experimental approach in the studies described above was different. In the salmonid studies (Alsop and Wood 1997; this study), $\dot{M}O_2$ and swimming performance were measured in step increment tests where each step was maintained for up to 20–40 min. However, the performance of cod (Blaikie and Kerr 1996) was measured in endurance tests where they were forced to swim at a fixed velocity for up to 8 d, while the U_{crit} of fed Atlantic cod was never measured. Therefore, it is not known whether the U_{crit} of Atlantic cod is reduced postprandially. Similarly, largemouth bass maintained their HI increment while swimming (Beamish 1974), but it is not known whether maximum swimming performance is reduced postprandially. Nonetheless, the general picture that emerges from these studies is that fish prioritize the metabolic processes associated with digestion and assimilation, and in some cases, this may be at the expense of maximum sustained swimming performance. Similarly, both varanid lizards and boa constrictor maintain the HI during activity (Table 2). However, the $\dot{V}O_{2max}$ of these reptiles is high enough to accommodate both oxygen demands of the HI and activity simultaneously (Table 2). In contrast, a limit on cardiovascular oxygen delivery could explain the decrease in U_{crit} of fed fish compared with starved fish.

Acknowledgments

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Literature Cited

- Alsop D.H. and C.W. Wood. 1997. The interactive effects of feeding and exercise on oxygen consumption, swimming performance and protein usage in juvenile rainbow trout (*Oncorhynchus mykiss*). *J Exp Biol* 200:2337–2346.
- Axelsson M., W.R. Driezic, A.P. Farrell, and S. Nilsson. 1989. Regulation of cardiac output and gut blood flow in the sea raven, *Hemitripterus americanus*. *Fish Physiol Biochem* 6: 315–326.
- Axelsson M. and R. Fritsche. 1991. Effects of exercise and feeding on the gastrointestinal blood flow in the Atlantic cod, *Gadus morhua*. *J Exp Biol* 158:181–198.
- Axelsson M., H. Thorarensen, S. Nilsson, and A.P. Farrell. 2000. Gastrointestinal blood flow in the red Irish lord, *Hemilepidotus hemilepidotus*: long-term effects of feeding and adrenergic control. *J Comp Physiol B* 170:145–152.
- Beamish F.W.H. 1974. The apparent specific dynamic action of largemouth bass, *Micropterus salmonides*. *J Fish Res Board Can* 31:1763–1769.
- Beamish F.W.H. and E.A. Trippel. 1990. Heat increment: a static or dynamic dimension in bioenergetic models? *Trans Am Fish Soc* 119:649–661.
- Bennett A.F. and J.W. Hicks. 2001. Postprandial exercise: prioritisation or additivity of the metabolic responses? *J Exp Biol* 204:2127–2132.
- Blaikie H.B. and S.R. Kerr. 1996. Effect of activity level on apparent heat increment in Atlantic cod, *Gadus morhua*. *Can J Fish Aquat Sci* 53:2093–2099.
- Brett J.R. and C.A. Zala. 1975. Daily pattern of nitrogen excretion and oxygen consumption of sockeye salmon (*Oncorhynchus nerka*) under controlled conditions. *J Fish Res Board Can* 32:2479–2486.
- Brown J.R. and J.N. Cameron. 1991. The relation between specific dynamic action and protein synthesis rates in channel catfish. *Physiol Zool* 64:298–309.
- Coulson R.A., J.D. Herbert, and T. Hernandez. 1978. Energy for amino acid absorption, transport and protein synthesis in vivo. *Comp Biochem Physiol* 60A:13–20.
- Furnell D.J. 1987. Partitioning of locomotor and feeding metabolism in sablefish (*Anoplopoma fimbria*). *Can J Zool* 65: 486–489.
- Gallaugh P.E., H. Thorarensen, and A.P. Farrell. 1995. Hematocrit and oxygen transport and swimming in rainbow trout (*Oncorhynchus mykiss*). *Respir Physiol* 102:279–292.
- Gallaugh P.E., H. Thorarensen, A. Kiessling, and A.P. Farrell. 2001. Effects of high intensity exercise training on cardiovascular function, oxygen uptake, internal oxygen transport and osmotic balance in chinook salmon (*Oncorhynchus tshawytscha*) during critical speed swimming. *J Exp Biol* 204: 2861–2872.
- Jobling M. and P.S. Davies. 1980. Effects of feeding on the

- metabolic rate and the specific dynamic action in plaice, *Pleuronectes platessa*. J Fish Biol 16:539–546.
- Kiceniuk J. and D.R. Jones. 1977. The oxygen transport system in trout (*Salmo gairdneri*) during sustained exercise. J Exp Biol 69:247–260.
- Legrow S.M. and F.W.H. Beamish. 1986. Influence of dietary protein and lipid on apparent heat increment of rainbow trout, *Salmo gairdneri*. Can J Fish Aquat Sci 43:19–25.
- Lyndon A.R., D.F. Houlihan, and S.J. Hall. 1992. The effect of short-term fasting and a single meal on protein synthesis and oxygen consumption in cod, *Gadus morhua*. J Comp Physiol B 162:209–215.
- McMillan D.N. and D.R. Houlihan. 1988. The effect of re-feeding on tissue protein synthesis in rainbow trout. Physiol Zool 61:429–441.
- Muir B.S. and A.J. Niimi. 1972. Oxygen consumption of the euryhaline fish Aholehole (*Kuhlia sandivicensis*) with reference to salinity, swimming, and food consumption. J Fish Res Board Can 29:67–77.
- Randall D.J. and C. Daxboeck. 1982. Cardiovascular changes in rainbow trout (*Salmo gairdneri*) during exercise. Can J Zool 60:1135–1140.
- Randall D.J. and P.A. Wright. 1987. Ammonia distribution and excretion in fish. Fish Physiol Biochem 3:107–120.
- Reidy S.P., S.R. Kerr, and J.A. Nelson. 2000. Aerobic and anaerobic swimming performance of individual cod. J Exp Biol 203:347–357.
- Saunders R.L. 1963. Respiration of the Atlantic cod. J Fish Res Board Can 20:373–386.
- Secor S.M., J.W. Hicks, and A.F. Bennett. 2000. Ventilatory and cardiovascular responses of a python (*Python molurus*) to exercise and digestion. J Exp Biol 203:2447–2454.
- Soofiani N.M. and I.G. Priede. 1985. Aerobic metabolic scope and swimming performance in juvenile cod, *Gadus morhua*. J Fish Biol 26:127–138.
- Tandler A. and F.W.H. Beamish. 1979. Mechanical and biochemical components of apparent specific dynamic action largemouth bass, *Micropterus salmoides* Lacepede. J Fish Biol 14:343–350.
- Thorarensen H. 1994. Gastrointestinal Blood Flow in Chinook Salmon *Oncorhynchus tshawytscha*. PhD thesis. Simon Fraser University, Burnaby.
- Thorarensen H., P.E. Gallagher, and A.P. Farrell. 1996. Cardiac output in swimming rainbow trout, *Oncorhynchus mykiss*, acclimated to seawater. Physiol Zool 69:139–153.
- Thorarensen H., P.E. Gallagher, A.K. Kiessling, and A.P. Farrell. 1993. Intestinal blood flow in swimming chinook salmon *Oncorhynchus tshawytscha* and the effects of haematocrit on blood flow distribution. J Exp Biol 179:115–129.